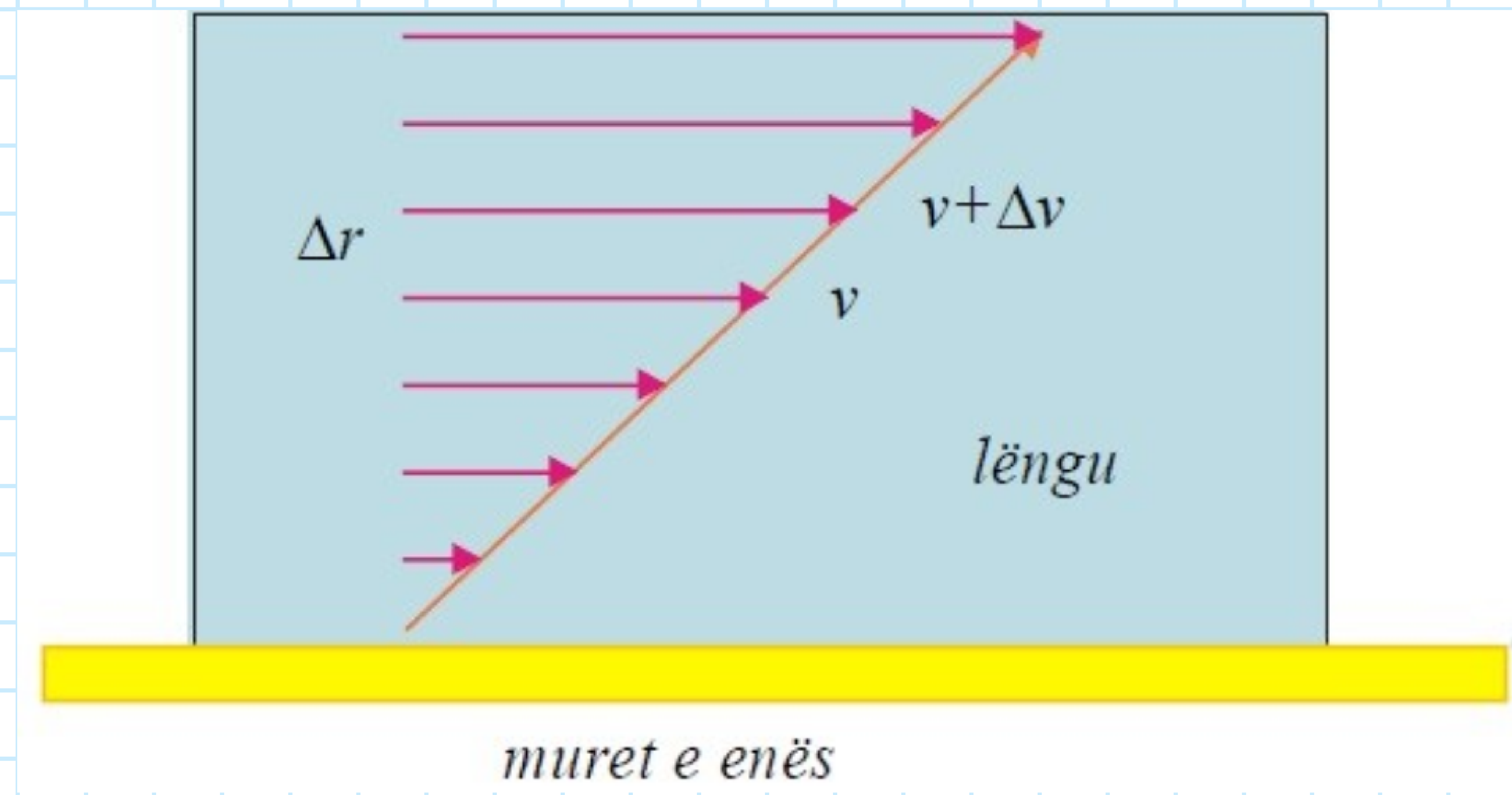
Ligjerata 8ë

Viskoziteti i fluideve. Formula e Stoksit

$$F \propto \frac{\Delta v}{\Delta r} \cdot S$$

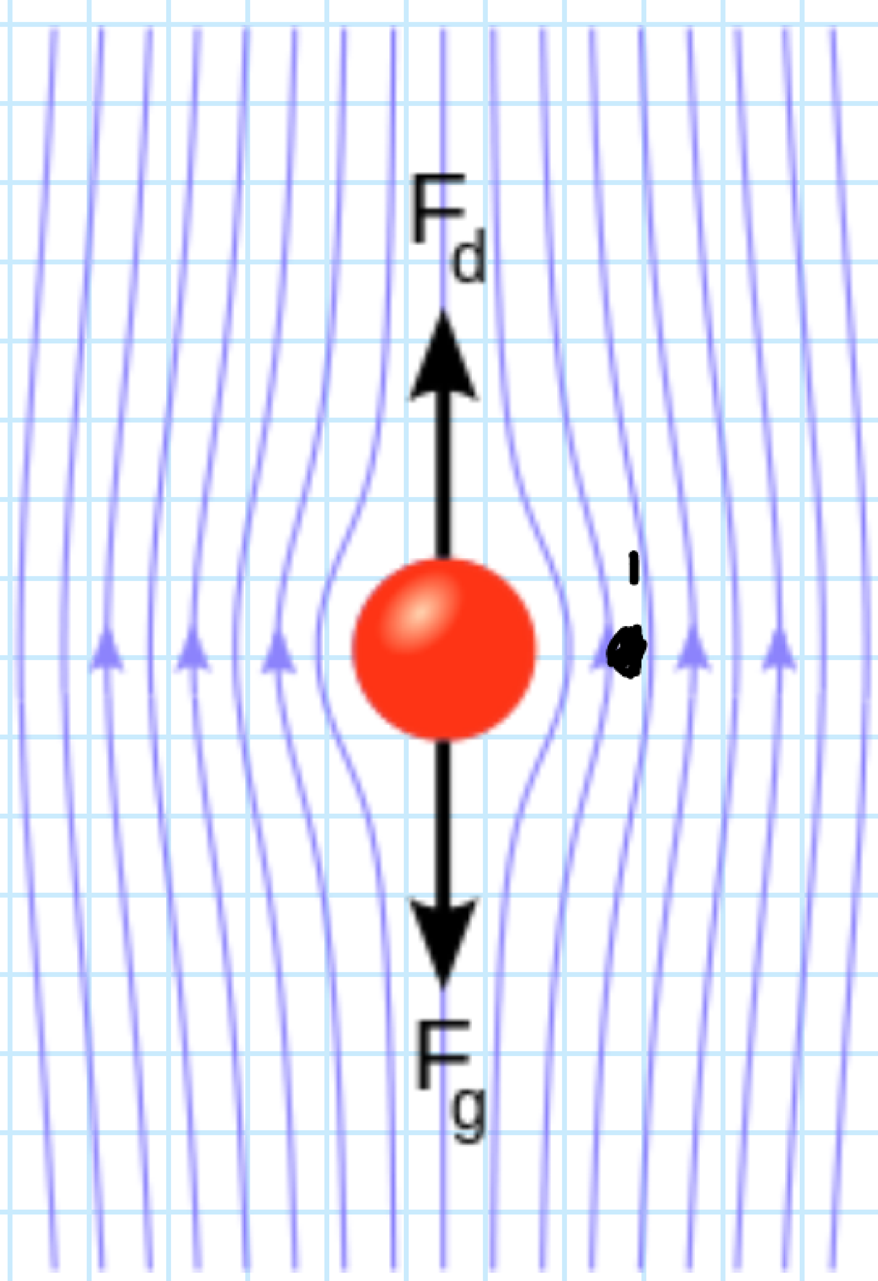


$$F = \eta \frac{\Delta v}{\Delta r} \cdot S$$

$$grad\, v = \frac{\Delta v}{\Delta r}$$

$$\eta = \frac{F}{\frac{\Delta v}{\Delta r} \cdot S} = \frac{N}{\frac{\cancel{m}}{s} \cdot \frac{1}{\cancel{m}} \cdot m^2} = \frac{Pa}{\frac{1}{s}} = \underline{Pa \cdot s}$$

$$\boxed{F_s = 6\pi \eta r v \text{ – vlen për sfera}}$$



$$F_s = F_g - F' = mg - m'g$$

$$= \rho V g - \rho' V \cdot g$$

$$\boxed{F_s = gV(\rho - \rho')}$$

$$6\pi \eta r v = gV(\rho - \rho') \; / \cdot 6\pi \eta r \qquad V = \frac{4}{3}\pi r^3$$

$$v = \frac{gV(\rho - \rho')}{6\pi \eta r} = \frac{g \frac{4}{3} \cancel{\pi} r^{\cancel{3}2} (\rho - \rho')}{6 \cancel{\pi} \eta \cancel{r}}$$

$$\boxed{v = \frac{2}{9} \cdot \frac{g(\rho - \rho') r^2}{\eta}}$$

$$\boxed{\eta = \frac{2}{9} \cdot \frac{g(\rho - \rho') \cdot r^2}{v}}$$

Formula e Stoksit

$$S = \pi r^2$$

$$F \propto p_d S, \quad p_d = \frac{1}{2} \rho v^2$$

$$F = k \frac{1}{2} \rho v^2 \pi r^2$$

$$k = \frac{12}{Re} = \frac{12}{\frac{\rho v r}{\eta}} = \frac{12\eta}{\rho v r}$$

$$F = \frac{{}^{6}\cancel{12}\eta}{\cancel{\rho v r}} \cdot \frac{1}{\cancel{2}} \cancel{\rho} v^{\cancel{2}} \pi r^{\cancel{2}} = 6\eta \pi v r$$